

Introduction To Algorithms

Week 7 - Similarity, Ranking, and Hashing
"Small Algorithms Under Data and AI Systems"

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Terms to Remember

Term	Brief Meaning
Unsupervised learning	Finding patterns without provided labels
Data mining lifecycle	Workflow for turning data into usable insight
Representation	How information is stored or described
Similarity	How alike two items are under a rule
Recommendation engine	System that suggests items
Content-based	Recommends from item features
Collaborative filtering	Recommends from user behavior patterns
Hybrid recommender	Combines multiple approaches
Cold start	Not enough starting data
Data sparsity	Too many missing or thin data points
Cipher	Method for disguising information
Plaintext	Original readable message
Ciphertext	Encoded or encrypted message
Cryptanalysis	Studying how to break or analyze ciphers
Hashing	Producing a stable fingerprint-like value
Decision tree	Branching model based on feature tests

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What We Will Use Today

- ▶ Data representation
- ▶ Similarity scoring
- ▶ Ranking tables
- ▶ Simple recommendation logic
- ▶ Assumptions and limitations
- ▶ Hashing for identity or integrity
- ▶ AI-assisted critique after manual framing

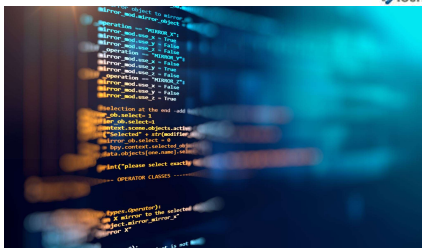
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What We Will "Touch On" and Revisit Later

- ▶ Full machine-learning workflows
- ▶ Larger datasets
- ▶ Model training
- ▶ Collaborative filtering at scale
- ▶ Cryptography beyond simple ciphers and hashing
- ▶ Decision trees and supervised learning in more depth
- ▶ Security, privacy, and responsible AI/data practice



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Review Coding Lab

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Review: What Lab 06 Taught Us

Last week - Representation shaped the result

What the nodes represented
What the edges represented
Where traversal started
What the model included
What the model left out

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Today's Question

How can a small algorithm compare, rank, recommend, group, or verify data?

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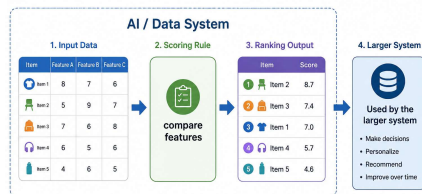
Small Algorithms Under Larger Systems

Larger systems often rely on small algorithmic choices:

- ▶ represent the data
- ▶ compare items
- ▶ score matches
- ▶ rank results
- ▶ group similar items
- ▶ check identity or integrity



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Small, clear algorithmic steps work together inside the system to produce useful results.

Algorithms in Systems

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What Counts as Success Today

make one AI/data idea visible

explain the representation

show evidence

name assumptions

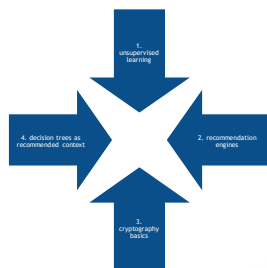
name a limitation

avoid overstating the result

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Textbook Review

Four Reading Areas



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Textbook Review

Unsupervised Learning

Unsupervised learning looks for patterns without provided answer labels.

Examples:

- ▶ grouping similar items
- ▶ finding clusters
- ▶ detecting structure
- ▶ discovering patterns

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Data Mining Lifecycle

1. business understanding
2. data understanding
3. data preparation
4. modeling
5. evaluation
6. deployment



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Data Mining Lifecycle



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Recommendation Engines

Recommendation engines suggest options by using data.

Common types:

- ▶ content-based
- ▶ collaborative filtering
- ▶ hybrid

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Cryptography Basics

The cryptography reading introduces:

- ▶ weakest link
- ▶ security goals
- ▶ sensitive data
- ▶ ciphers
- ▶ plaintext and ciphertext
- ▶ cryptanalysis

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Decision Trees

Recommended reading introduces decision trees.

For this week:

- ▶ recognize the branching idea
- ▶ notice strengths and weaknesses
- ▶ save deeper use for later AI/data courses

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Deeply Learn

Deeply learn now:

- ▶ representation
- ▶ similarity
- ▶ ranking
- ▶ assumptions
- ▶ limitations
- ▶ visible evidence

Recognize for later:

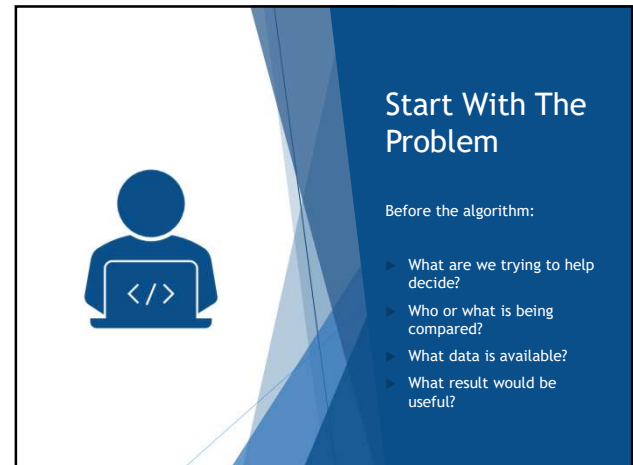
- ▶ full ML workflows
- ▶ advanced recommendation systems
- ▶ full cryptography
- ▶ decision-tree modeling



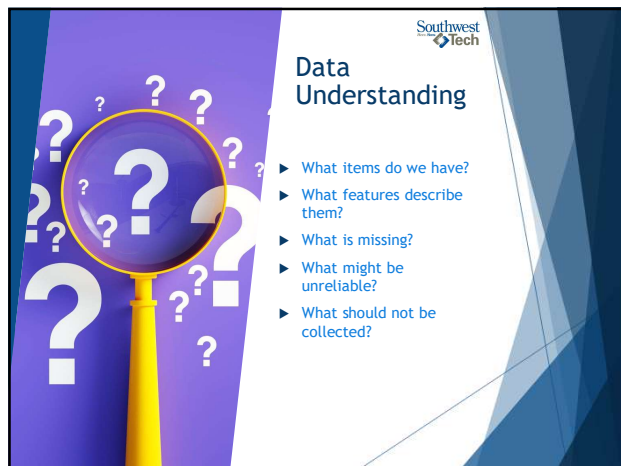
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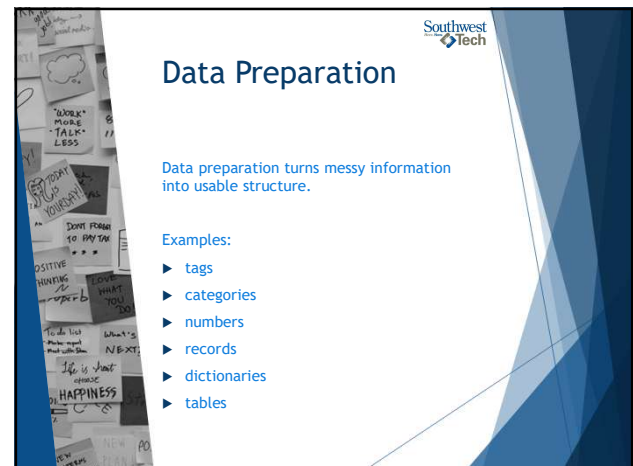
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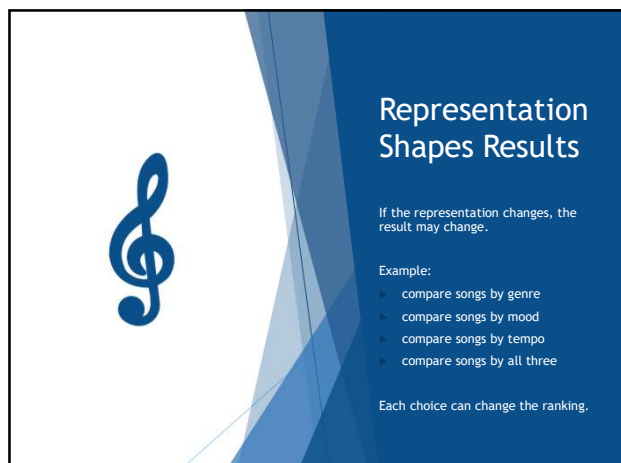
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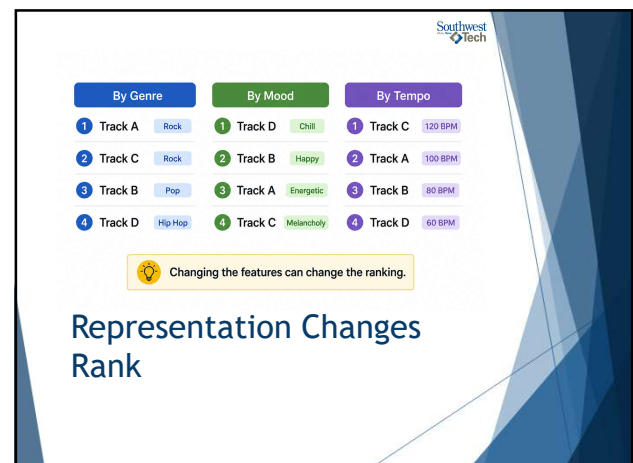
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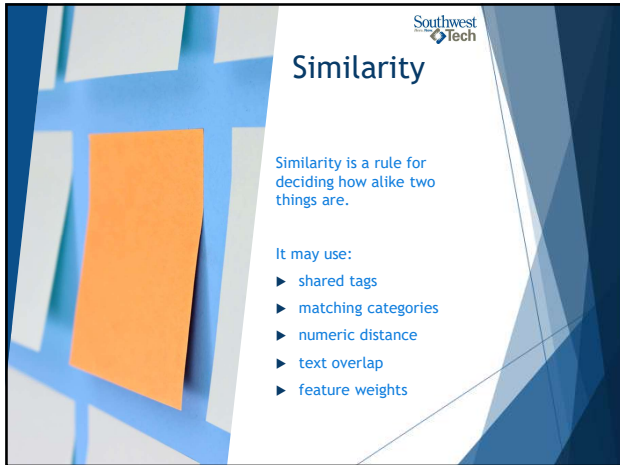
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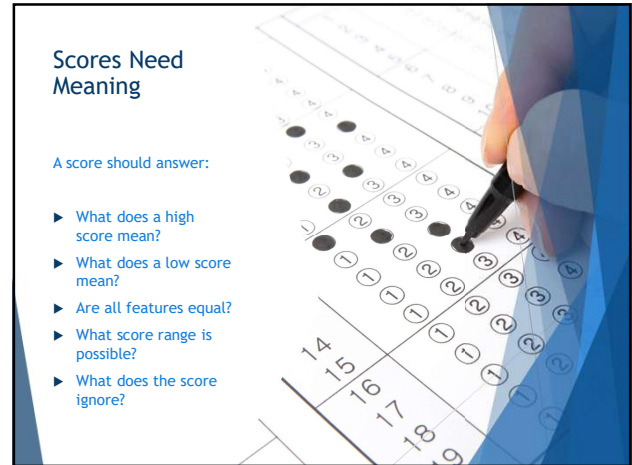
Similarity

Similarity is a rule for deciding how alike two things are.

It may use:

- ▶ shared tags
- ▶ matching categories
- ▶ numeric distance
- ▶ text overlap
- ▶ feature weights

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Scores Need Meaning

A score should answer:

- ▶ What does a high score mean?
- ▶ What does a low score mean?
- ▶ Are all features equal?
- ▶ What score range is possible?
- ▶ What does the score ignore?

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Similarity, Ranking, and Recommendation

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Ranking

Ranking orders items by a score or rule.

Ranking can show:

- ▶ strongest match
- ▶ weakest match
- ▶ top recommendation
- ▶ items needing review

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Ranking

Recommended Resources

	Item	Score
1	Practice Set B	92
2	Video Walkthrough A	87
3	Reference Sheet C	74
4	Challenge Task D	61

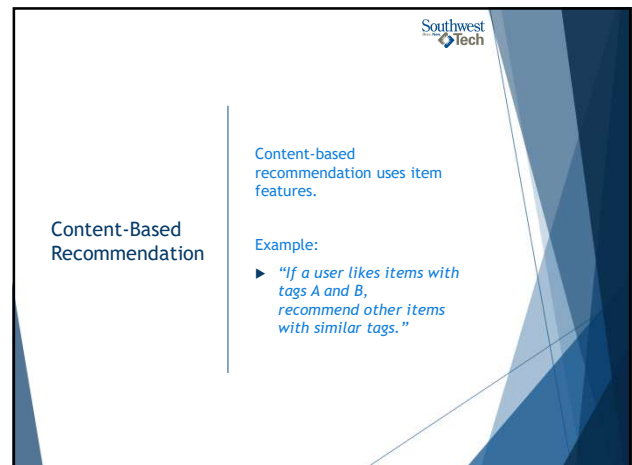
Scoring Rule (Transparent)

topic match + difficulty fit + recent progress

Score = topic match + difficulty fit + recent progress

The top item is highest by this rule, not universally best.

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Content-Based Recommendation

Content-based recommendation uses item features.

Example:

- ▶ "If a user likes items with tags A and B, recommend other items with similar tags."

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Similarity In Unstructured Documents

Text can be compared by features such as:

- ▶ words
- ▶ tags
- ▶ topics
- ▶ categories
- ▶ extracted terms

The representation still matters.

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Collaborative Filtering



Collaborative filtering uses patterns from users or behavior.

It may ask:

- ▶ What did similar users choose?
- ▶ What did people with similar histories like?
- ▶ What patterns appear across many users?

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Collaborative Filtering Risks



Risks include:

- ▶ limited sample size
- ▶ over-reliance on history
- ▶ isolated analysis
- ▶ reinforcing past patterns
- ▶ weak recommendations for new users

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Hybrid Recommendation

Hybrid recommendation combines signals.

It may use:

- ▶ item features
- ▶ user history
- ▶ similarity scores
- ▶ reference vectors
- ▶ feedback over time

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Similarity Matrix And Reference Vector

Two useful structures:

- ▶ similarity matrix: compares items to items
- ▶ reference vector: represents a user, target, or profile

Both make comparison visible.

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Similarity Matrix And Reference Vector

	Item A	Item B	Item C	Item D
Item A	1.0	0.8	0.3	0.6
Item B	0.8	1.0	0.4	0.7
Item C	0.3	0.4	1.0	0.5
Item D	0.6	0.7	0.5	1.0

compare patterns

Reference Vector	
Genre match	0.8
Mood match	0.6
Tempo match	0.4
Difficulty fit	0.7

Each item's pattern is compared to the user's reference vector.

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Recommendation Limits

Recommendation systems have limits:

- ▶ cold start
- ▶ metadata requirements
- ▶ data sparsity
- ▶ social influence
- ▶ missing context
- ▶ biased or incomplete data

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Recommendation Limits

Recommendations are useful, but they depend on available data.

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Cold Start And Sparse Data

Cold start:

- ▶ not enough information at the beginning

Data sparsity:

- ▶ too many blanks or thin signals

Both weaken recommendations.

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Social Influence Is Double-Edged

Social influence can:

- ▶ reveal useful patterns
- ▶ amplify popularity
- ▶ hide better-fit options
- ▶ reinforce existing bias
- ▶ make recommendations feel objective

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Clustering As Grouping By Closeness

Clustering groups items by closeness or similarity.

(For Lab 07), Clustering can be simple:

- ▶ choose features
- ▶ compare items
- ▶ create groups
- ▶ explain why the groups make sense
- ▶ name what the grouping does not prove

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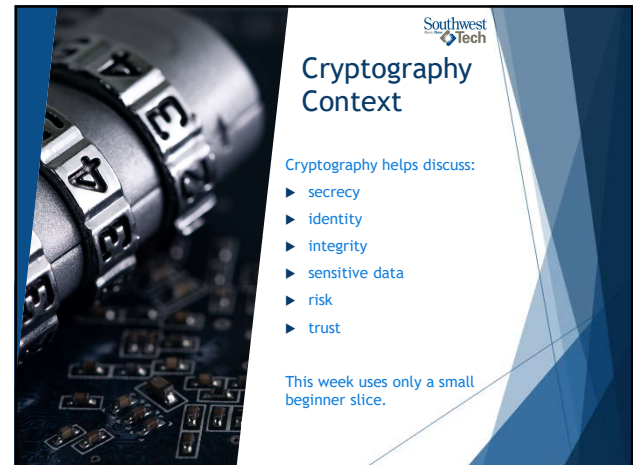
Break

10 Minute "Human" bio-break

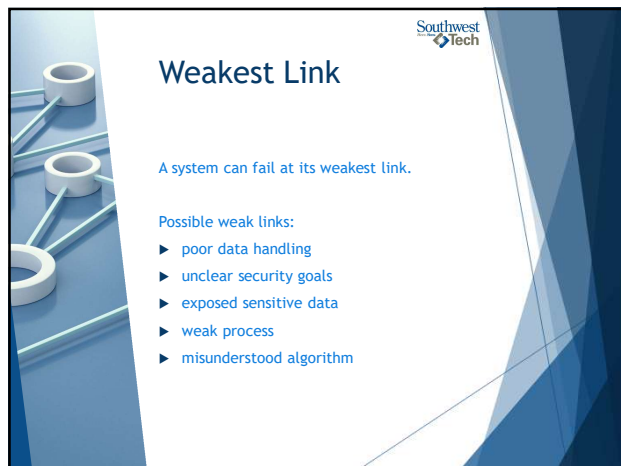
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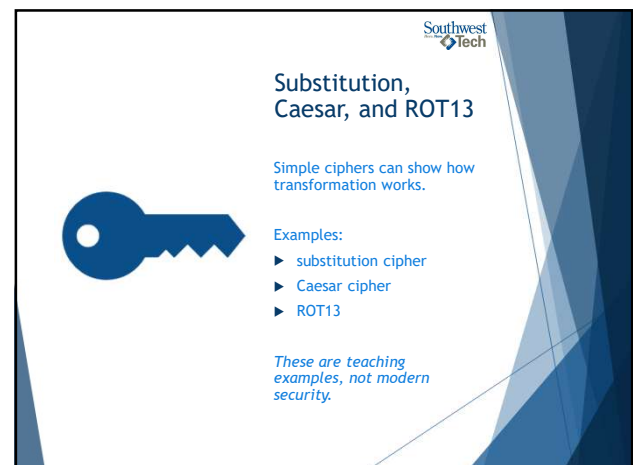
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Cryptanalysis And Transposition



Cryptanalysis asks how a method might be analyzed or broken.



Transposition ciphers rearrange positions rather than substituting symbols.



Both show that algorithm design includes attack thinking.

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Hashing



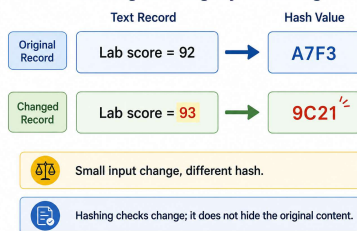
Hashing produces a stable value from input data.

Useful for:

- ▶ identity checks
- ▶ integrity checks
- ▶ lookup support
- ▶ change detection

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Hashing as Integrity Checking



Hashing

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Hashing vs Encryption

Hashing:

- ▶ one-way check value
- ▶ useful for identity or integrity

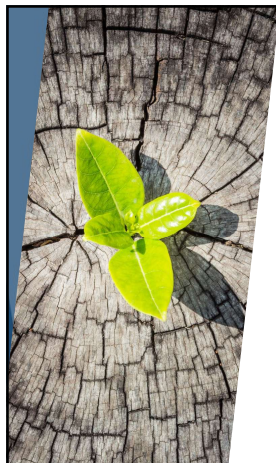
Encryption:

- ▶ protects readable content
- ▶ should be decryptable by authorized parties

Do not treat them as the same.

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Decision Tree Recognition



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Decision Tree Recognition

A decision tree uses branching questions.

Example shape:

- ▶ Is the feature present?
- ▶ Is the value above a threshold?
- ▶ Which branch applies?
- ▶ What class or decision results?

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Decision Tree Strengths And Weaknesses

Strengths

- ▶ explainable structure
- ▶ visible decision path
- ▶ useful for classification

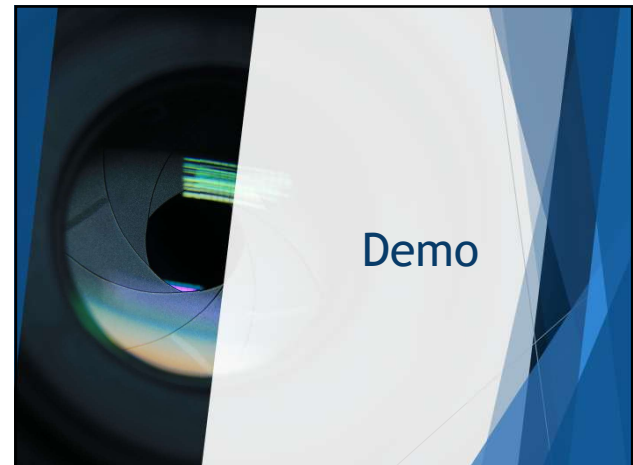


Weaknesses

- ▶ can overfit
- ▶ depends on selected features
- ▶ can become brittle



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Demo Scenario

Problem:

Create a tiny music recommendation example.

- Represent songs with simple tags such as genre, mood, tempo, and instrument.
- Compare one reference song to several candidate songs using a basic similarity score based on shared tags.

Goal

Use:

- a reference song
- candidate songs
- simple tags
- tag-overlap similarity
- ranked recommendations

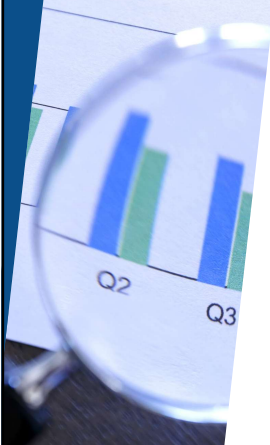
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Demo Evidence

The demo produces:

- ▶ item-feature table
- ▶ reference song summary
- ▶ similarity score table
- ▶ ranked recommendation
- ▶ assumption and limitation notes



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Demo Key Questions



- ▶ Which tags are being compared?
- ▶ What does the score mean?
- ▶ Which feature is missing?
- ▶ Why did the top item win?
- ▶ What should the result not claim?

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Presentation Layer Progression

Same logic, different presentation:

1. plain tables
2. ANSI color
3. Rich tables and panels



The algorithm should not change.

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Console Presentation Styles

Plain

```

Scan started...
File: data.csv
Size: 2.1 MB
Status: PASS
Issues found: 0
File: report.txt
Size: 18 KB
Status: FAIL
Issues found: 2
File: notes.log
Size: 124 KB
Status: FOUND
Issues found: 5
Scan complete.

```

ANSI Color

```

Scan started...
File: data.csv
Size: 2.1 MB
Status: PASS
Issues found: 0
File: report.txt
Size: 18 KB
Status: FAIL
Issues found: 2
File: notes.log
Size: 124 KB
Status: FOUND
Issues found: 5
Scan complete.

```

Rich

File	Size	Status	Issues
data.csv	2.1 MB	PASS	0
report.txt	18 KB	FAIL	2
notes.log	124 KB	FOUND	5

Scan complete.

💡 Presentation layer can improve readability without changing the algorithm.

Presentation Styles

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Demo Explanation

"The top recommendation ranked highest because it shared the most tags with the reference song. This result depends on the tags I chose, and it does not prove the listener would actually prefer the song."

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From Demo to Lab

In Lab 07, choose or receive one option:

- similarity ranking
- simple clustering
- tiny recommendation
- hashing demonstration

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Lab 07 Evidence

similarity matrix

ranking table

cluster grouping table

scatter plot

hash comparison

representation table

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Four Acceptable Forms of Lab Evidence

1 Ranking Table

Item	Score
1. Practice Set B	92
2. Video Walkthrough A	87
3. Reference Sheet C	74
4. Challenge Task D	61

2 Cluster Table

Cluster	Items
Cluster 1	Item A, Item B, Item C
Cluster 2	Item D, Item E, Item F

3 Hash Comparison

Record	Hash Value
Original Record	A7F3
Changed Record	9C21

4 Representation Table

Item	Genre	Mood	Tempo	Difficulty
Item A	Rock	Energetic	Fast	Medium
Item B	Pop	Happy	Medium	Easy
Item C	Jazz	Calm	Slow	Hard
Item D	Hip Hop	Focused	Medium	Medium

💡 Evidence should make the reasoning visible.

Evidence Choices

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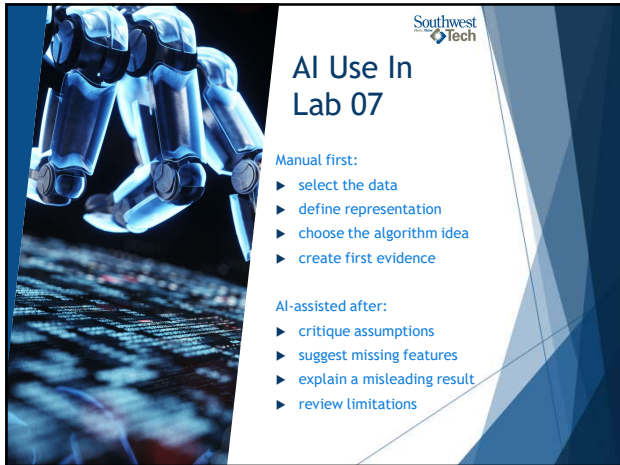
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README Evidence

Your README should show:

- selected option
- data set
- representation explanation
- visible evidence
- two assumptions
- one limitation or risk
- AI/data connection
- AI-use note, if applicable

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AI Use In Lab 07

Manual first:

- ▶ select the data
- ▶ define representation
- ▶ choose the algorithm idea
- ▶ create first evidence

AI-assisted after:

- ▶ critique assumptions
- ▶ suggest missing features
- ▶ explain a misleading result
- ▶ review limitations

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Wrap up

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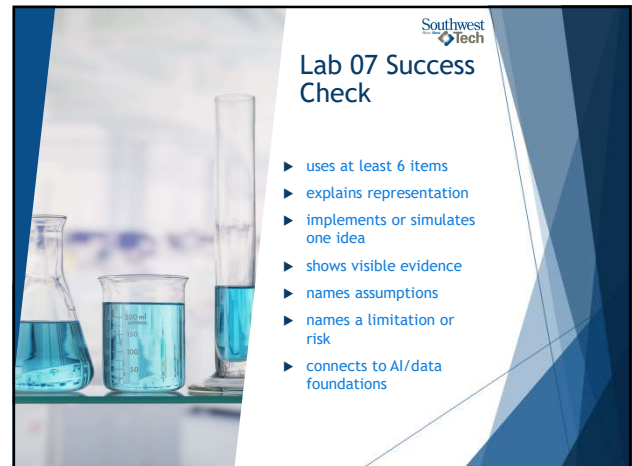
What To Carry Forward



Carry forward:

- ▶ representation shapes results
- ▶ similarity is a design choice
- ▶ rankings need explanation
- ▶ small data has limits
- ▶ hashing can support integrity
- ▶ evidence must be visible

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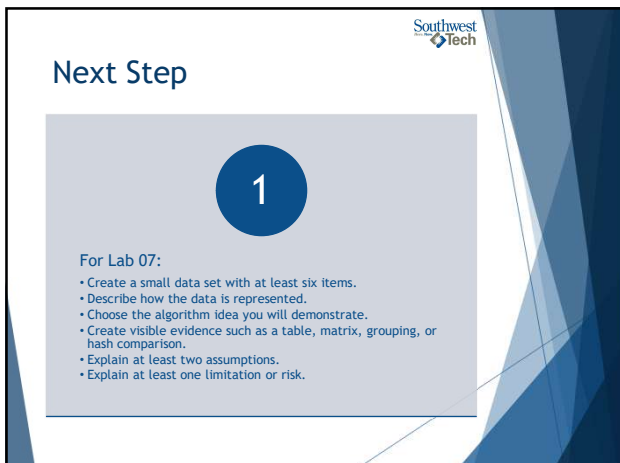
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Lab 07 Success Check



- ▶ uses at least 6 items
- ▶ explains representation
- ▶ implements or simulates one idea
- ▶ shows visible evidence
- ▶ names assumptions
- ▶ names a limitation or risk
- ▶ connects to AI/data foundations

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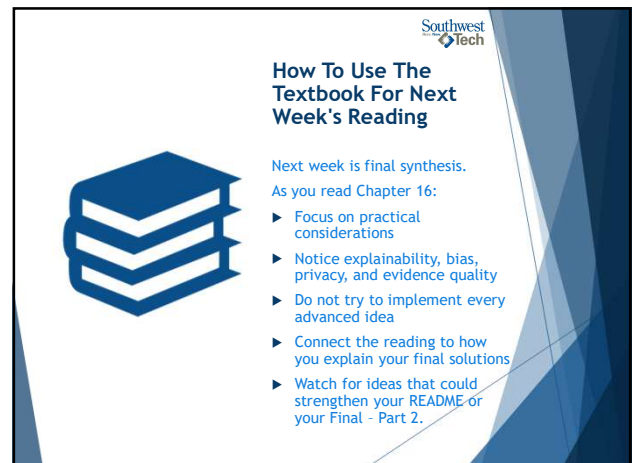
Next Step

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For Lab 07:

- Create a small data set with at least six items.
- Describe how the data is represented.
- Choose the algorithm idea you will demonstrate.
- Create visible evidence such as a table, matrix, grouping, or hash comparison.
- Explain at least two assumptions.
- Explain at least one limitation or risk.

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How To Use The Textbook For Next Week's Reading



Next week is final synthesis.

As you read Chapter 16:

- ▶ Focus on practical considerations
- ▶ Notice explainability, bias, privacy, and evidence quality
- ▶ Do not try to implement every advanced idea
- ▶ Connect the reading to how you explain your final solutions
- ▶ Watch for ideas that could strengthen your README or your Final - Part 2.

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